STUDYFLOW

Test plan and verification results

Automated tests, Android screen flow and manual acceptance scenarios

Report 03 / Version 1.0.0+1 / November 4, 2026

Run summary
November 4, 2026: all 8 existing unit and widget tests passed. Code analysis was clean. Android screenshot capture ran separately. The manual scenarios are not all treated as executed; expected outcomes and verification status are stated below.

This document covers working examples and failure conditions. It uses repeatable steps to examine data loss, repeated requests, account authorization and language changes alongside the successful workflow.

Scope	Contents
Automated tests	A01-A08: existing source tests and results from the recorded run.
Android visual workflow	G01: emulator navigation and capture with a demonstration plan.
Manual acceptance list	M01-M45: prerequisites, actions and expected results.
Status definitions	Passed: an executed automated check. Observed: visible in the screen workflow. Pending: separate manual verification is required.

Test runner machine output success: true. The final event time is 39.4 seconds. This is the runner’s event time, not an application performance measurement.

Automated tests / data and language

A01 / Reopen the local plan

Procedure: Create a Mobile Programming course, a high-priority task and a 25-minute session. Complete the task, then close and reopen the Hive box.

Expected: Completion, priority and session minutes persist. Deleting the course removes its tasks but preserves the session.

Result: Passed. Source: test/widget_test.dart

A02 / Protect corrupt storage

Procedure: Write invalid JSON under data. Load the ViewModel and attempt to add a course.

Expected: ready is false; adding returns false and the corrupt content is unchanged.

Result: Passed. Source: test/widget_test.dart

A03 / Persist the language preference

Procedure: Select en using LanguageController, then close and reopen the Hive box.

Expected: A new controller reads the en code.

Result: Passed. Source: test/widget_test.dart

A04 / Language, date picker and user text

Procedure: Select English at 390 × 844. Observe the demonstration course named “Ders”; open a new task and date picker, then rebuild the app widget.

Expected: Overview and New task are displayed; the course name is unchanged; the date picker uses en and the preference survives reopening.

Result: Passed. Source: test/widget_test.dart

Automated tests / workflow and errors

A05 / Local workflow on a narrow screen

Procedure: Create a course and task, complete the task and find it under Done. Finish a 15-minute timer using a controlled clock and save the session.

Expected: The task is visible; saving a session and navigating to Progress finish without errors.

Result: Passed. Source: test/widget_test.dart

A06 / Empty authDomain correction

Procedure: Read CloudService.options.authDomain when the build variable is empty.

Expected: The Android SDK receives null rather than an empty string.

Result: Passed. Source: test/cloud_ui_test.dart

A07 / Loading and recovery after an error

Procedure: FakeCloud delays initialization and registration. Finish initialization, open the form and complete registration with email-already-in-use.

Expected: The spinner is visible while pending; back navigation and inputs are locked; one request is made. After the error the spinner closes, inputs unlock, email is retained and a helpful message appears.

Result: Passed. Source: test/cloud_ui_test.dart

A08 / Hide raw error details

Procedure: Prepare an example FirebaseAuthException with internal-error and private content in its message.

Expected: The user message includes the error code but excludes the raw private details.

Result: Passed. Source: test/cloud_ui_test.dart

Android screen workflow and method

G01 / Emulator screenshot capture

The real widget tree runs on an Android 17 / API 37 emulator. A separate study_flow_report box is populated with three courses, four tasks and seven focus sessions. Overview, Tasks, New task, Focus, Progress and the language menu are opened. Overview is captured in English, then the app returns to Turkish. Account, registration, loading and error states use controlled service responses. For this English edition, report_capture_en_test.dart repeats the capture in English using a separate study_flow_report_en box and English demonstration names.

Criterion	Expectation / observation
Data separation	The normal study_flow box and real Firebase data are not modified.
Interface	Main tabs and the new-task form are accessible; both language interfaces have been captured.
Loading	A spinner appears before completion; the form returns after an error response.
Visual output	Real Android PNGs are saved under teslim/ekranlar; English captures under teslim/screens_en.
Outside scope	This flow does not validate real account creation, production Firestore rules or remote backup content.

Repeat the original capture

```
flutter drive --driver=test_driver/report_driver.dart
  --target=integration_test/report_capture_test.dart
  -d emulator-5554 --dart-define-from-file=firebase.local.json
```

Separation of test environments

A01-A03 use real temporary Hive files. Widget tests use an in-memory repository so disk latency does not affect the test clock. FakeCloud in A07 controls initialization and registration with Completer, allowing loading and failure checks without creating a real account.

Unless stated otherwise, manual scenarios require the app to be open on a test device and a backed-up demonstration plan to be ready. Run deletion and restore cases on test data rather than personal records.

TEST SCENARIOS

Course and task management

These are manual acceptance scenarios. Pending means that a separate manual result record was not created in the source report run. Overlapping automated behavior is evidenced separately under the relevant A identifiers.

ID / scenario	Procedure	Expected result
M01 **Add a course**	Overview → Add course. Choose a valid name and color, then Save.	The course is listed and remains after restarting the app. Status: Pending
M02 **Empty course name**	Save a blank or whitespace-only course name.	A validation message appears; no course is created. Status: Pending
M03 **Turkish user data**	Add a course named “Ölçme ve Değerlendirme” and a task named “Çalışma özeti”.	The characters ç, ğ, ı, ö, ş and ü are preserved in storage and after reopening. Status: Pending
M04 **Task without a course**	In an empty plan, select Tasks → Add.	The course creation dialog opens first. Status: Pending
M05 **Task form limits**	Save with an empty title, then test the 120-character limit.	An empty title is rejected; the field accepts no characters beyond the limit. Status: Pending
M06 **Priority and date sorting**	Add tasks with different priorities on the same date and tasks on different dates.	Due date sorts first; higher priority comes first on matching dates. Status: Pending
M07 **Search and filter**	Search part of a title and switch Pending / Done / All.	Only tasks matching both search and status conditions appear. Status: Pending
M08 **Deletion confirmation and relations**	Open task and course deletion; cancel first, then confirm.	Cancel preserves data. Deleting a course removes its tasks but preserves focus sessions. Status: Pending

Focus timer and progress

These are manual acceptance scenarios. Pending means that a separate manual result record was not created in the source report run. Overlapping automated behavior is evidenced separately under the relevant A identifiers.

ID / scenario	Procedure	Expected result
M09 **Duration selection**	Select 15, 25 and 45 minutes while the timer is idle.	Starting duration and ring reflect the selection. Status: Pending
M10 **Pause and resume**	Start the timer, pause it, wait and resume.	Remaining time is preserved while paused and continues from that value. Status: Pending
M11 **Change tabs**	Switch from a running timer to Tasks, then return to Focus.	The timer is not reset; elapsed time is deducted. Status: Pending
M12 **Background the app**	Move a running timer to the background and return.	Remaining time is recalculated from the actual end time. Status: Pending
M13 **Completion and saving**	Finish a session. Check the chart before saving, then save and check again.	Time is not added automatically; saving adds one session. Status: Pending
M14 **Reset and termination**	Reset the timer; in a separate trial, terminate the app completely.	Reset returns to the selected duration. Termination discards the active session; saved sessions remain. Status: Pending
M15 **Seven-day window**	Prepare sessions for today, within the seven-day period and outside it.	Only the last seven days count in the chart; daily totals are correct. Status: Pending
M16 **Course without tasks**	Create a course with no tasks and a course with completed tasks.	The empty course shows zero; the other shows completed / total tasks. Status: Pending

TEST SCENARIOS

Local data and language

These are manual acceptance scenarios. Pending means that a separate manual result record was not created in the source report run. Overlapping automated behavior is evidenced separately under the relevant A identifiers.

ID / scenario	Procedure	Expected result
M17 **Offline planning**	Disable networking; add and complete courses/tasks, then reopen the app.	The local plan and saved sessions remain usable. Status: Pending
M18 **Rollback after a save error**	Make the test repository fail a write and attempt a task change.	The previous plan is restored and a saving error is shown. Status: Pending
M19 **TR → EN → TR**	Change language in both directions and visit the main tabs and forms.	Headings, buttons, validation and loading messages follow the selected language. Status: Pending
M20 **Preserve user text**	Save “Ders” or “Görevler”, which resemble translation keys, as user data.	User-entered text is not translated when the interface language changes. Status: Pending
M21 **Language preference write failure**	Make the test repository fail the language write.	The previous language is restored and a saving error is reported. Status: Pending
M22 **Local and cloud preferences**	Restore another plan backup while English is selected.	The plan changes; the device’s language preference does not. Status: Pending
M23 **Device / browser separation**	Create different plans on Android and web.	Local records are not shared automatically; account-based backup transfer is required. Status: Pending

Accounts and error handling

These are manual acceptance scenarios. Pending means that a separate manual result record was not created in the source report run. Overlapping automated behavior is evidenced separately under the relevant A identifiers.

ID / scenario	Procedure	Expected result
M24 **New account**	With Email/Password enabled, register using a valid unused email and strong password.	A user appears in Firebase Auth and is signed in. Status: Pending
M25 **Existing account**	Attempt registration with an already registered email.	An email-in-use message suggests signing in. Status: Pending
M26 **Invalid sign-in**	Sign in with a wrong password or unknown account.	A sign-in error appears without sensitive details. Status: Pending
M27 **Invalid form**	Submit an email without @ or a password shorter than six characters.	Local validation fails and no server request starts. Status: Pending
M28 **Network and service failure**	Disconnect during sign-in; separately try a project with Email/Password disabled.	A helpful network or provider error appears and the form remains reusable. Status: Pending
M29 **Spinner and repeated clicks**	With a delayed response, press the account button repeatedly and attempt back navigation.	Only one request runs; the loading overlay blocks background interaction. Status: Pending
M30 **Session and sign-out**	Reopen after sign-in, then sign out.	The SDK preserves the session; sign-out returns to the form without deleting the local plan. Status: Pending

Cloud backup and access

These are manual acceptance scenarios. Pending means that a separate manual result record was not created in the source report run. Overlapping automated behavior is evidenced separately under the relevant A identifiers.

ID / scenario	Procedure	Expected result
M31 **First backup**	With a saved plan and active session, confirm cloud backup.	users/{uid}/backups/planner contains data with courses, tasks and sessions. Status: Pending
M32 **Update a backup**	Add a local task. Inspect Firestore before backing up, then back up and inspect again.	The old backup remains initially; confirmed upload replaces the complete plan. Status: Pending
M33 **Cancel upload**	Choose Cancel in backup confirmation.	The server plan is unchanged. Status: Pending
M34 **Restore a plan**	Prepare a different cloud plan and confirm restoration.	The local plan is replaced by server content and saved persistently. Status: Pending
M35 **Missing backup / cancel**	Restore with an account that has no backup; separately cancel restoration.	No-backup feedback or cancellation preserves the local plan. Status: Pending
M36 **Unauthenticated access**	Use Rules Playground or Emulator to read and write without authentication.	Requests are denied. Status: Pending
M37 **Another user’s data**	As user A, attempt to read and write user B’s backup path.	Both operations are denied. Status: Pending
M38 **Document structure and timestamp**	Try extra fields, non-map data and a client timestamp.	Invalid writes are rejected; a valid write with serverTimestamp is accepted. Status: Pending

Service, interface and distribution

These are manual acceptance scenarios. Pending means that a separate manual result record was not created in the source report run. Overlapping automated behavior is evidenced separately under the relevant A identifiers.

ID / scenario	Procedure	Expected result
M39 **Holiday list**	Open the calendar online and check name fields in each language.	Records match the year and country; tr uses localName and en uses name. Status: Pending
M40 **Holiday error / retry**	Simulate an HTTP failure or offline state, then reconnect and select Try again.	Error and retry feedback appear; a successful response loads the list. Status: Pending
M41 **Holiday timeout**	Use a controlled response delayed beyond 12 seconds.	Waiting ends, an error appears and the local plan is unaffected. Status: Pending
M42 **Small screen / keyboard**	Use task and account forms on a narrow screen with the keyboard visible.	Required fields and buttons are reachable by scrolling without overflow. Status: Pending
M43 **Large text and accessibility**	Increase system text scale; inspect buttons and loading with a screen reader.	Core actions remain readable and accessible; loading state is announced. Status: Pending
M44 **Clean installation / reinstallation**	Install the signed test APK on a clean test device and try basic planning.	The correct package opens. Account for local data removal when uninstalling and reinstalling. Status: Pending
M45 **Physical device / web**	Repeat the local workflow on a physical Android device and web with platform-specific settings.	The layout and core features work; Android configuration is not used instead of a web app ID. Status: Pending

Acceptance criteria and result records

Pre-release priorities

Persistence, corrupt-record protection, course-task relationships and loading locks are core acceptance criteria. If cloud features are used, complete M24, M31, M34 and M36-M38 in a real Firebase test project. Physical-device checks and large-text testing must be performed separately from emulator screenshots.

Result record field	Information to record
Scenario identifier	For example, M34.
Environment	Device model, Android version, app version and network state.
Actual outcome	Whether it matches the expectation; describe any difference.
Evidence	Screenshot or error code without personal information.
Decision	Passed / Failed / Blocked; retest date.

Known behavior

Losing an active timer when the app closes completely, not caching holiday data offline and replacing the entire plan on cloud restore are defined behaviors of this version. Compare them with the expected scope before reporting them as defects.

Verifiable sources

Automated scenarios: test/widget_test.dart and test/cloud_ui_test.dart. Screen workflow: integration_test/report_capture_test.dart. Image transfer: test_driver/report_driver.dart. Original raw run output: tmp/final_reports/test-results.jsonl. Security expectations: firestore.rules. English capture: integration_test/report_capture_en_test.dart and test_driver/report_en_driver.dart.